

Action Ring

Product Requirements Document — a customizable radial quick-action overlay, feature-complete clone of the Logitech Actions Ring (plus fixes for its known weaknesses)

Document status	Draft v1.0
Date	July 14, 2026
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Target platform	Windows 10/11 first; macOS 13+ second; Linux stretch goal
Research basis	3 parallel research reports: official Logitech documentation & release notes, ~25 press reviews, community threads (Reddit, forums, Hacker News) and 12+ competitor tools. Full reports in research/ .

1. Executive Summary

The Logitech **Actions Ring** (shipped in Logi Options+ v1.83, Oct 14 2024; extended to all MX devices Aug 2025 and to all compatible Logitech devices June 2026) is an on-screen radial overlay of **eight customizable "bubbles"** that appears around the mouse cursor when a trigger button is pressed. Each bubble executes an action — a keystroke, macro, app launch, media control, device switch, plugin command, or an analog "adjustment" driven by the scroll wheel — and the ring is **contextual**: it swaps its contents based on the foreground application. Reviewers call it "the best productivity mouse feature I've ever used" (MakeUseOf) and "a reason to buy the MX Master 4" by itself (AppleInsider). Logitech claims up to 33% time savings and 63% fewer repetitive mouse movements (Ergo Lab study, 2025, n=37).

This PRD specifies **Action Ring**, an independent application that replicates every documented feature of Logitech's implementation **without requiring Logitech hardware or an internet connection**, and fixes the seventeen most-requested community complaints (cold-start latency, no disable toggle, 8-slot cap, no offline mode, no multi-keystroke sequences, etc.). The document ends with a phased work breakdown (\$12) suitable for import into a task tracker.

2. Background & Feature Timeline (as shipped by Logitech)

Date	Options+ ver.	Milestone
Apr 17, 2024	—	Logi AI Prompt Builder launches (ChatGPT pop-up window, "recipes": Rephrase / Summarize / Reply / Create Email + custom). Not yet a ring, but later reachable from it via Smart Actions.
Sep 24 / Oct 14, 2024	1.83	Actions Ring debuts , exclusive to the MX Creative Console (trigger: Dialpad bottom-right button), alongside the Logi Marketplace plugin store.
Apr 21, 2025	1.89	"Open Folder" action; improved Adapt-to-App profile activation.
Jun 9, 2025	1.91	Folders (sub-groups of 9 sub-bubbles per bubble → up to 72 actions); improved Switch Profile action.
Aug 8, 2025	1.93	Extended to all MX mice and keyboards ; Easy-Switch available inside the ring.
Sep–Oct 2025	1.95	MX Master 4 ships with a dedicated haptic thumb-panel trigger; hover/select haptic bumps; ring opacity improved; Easy-Switch ×1/×2/×3 actions.
Dec 3, 2025	1.98	Plugin-service-off warning on trigger; per-action adjustment sensitivity; icon editor upgrades.
Feb 9, 2026	1.99	MX Ergo support.
Mar–Jun 2026	2.1–2.4	Profile import; Mac wake responsiveness fix; SVG icon compatibility popup; extended to all compatible Logitech devices ; ring display sizes Small / Medium / Large.

3. Goals and Non-Goals

Goals

- Ship a radial overlay functionally equivalent to Logitech's Actions Ring: 8 bubbles, folders, adjustment bars, contextual per-app profiles, drag-and-drop configuration, plugin actions.

2. Work with **any** mouse/keyboard (mappable trigger: any mouse button, keyboard hotkey, or gesture) — no hardware lock-in.
3. Be **offline-first**: zero internet requirement for any core feature; local JSON config; optional cloud sync only as opt-in.
4. Beat Logitech on the community's top complaints: instant invocation (<100 ms always, including first use after wake), a real enable/disable toggle + game mode, configurable slot count, multi-keystroke sequences, and light resource usage (<50 MB RAM idle, <0.1% CPU).

Non-Goals (v1)

1. Hardware haptic feedback (requires device firmware; consider Windows 11 Advanced Haptics API as a v2 stretch).
2. Cloud marketplace with paid content; v1 ships a curated local plugin/preset gallery instead.
3. ChatGPT-account-integrated AI Prompt Builder clone; v1 exposes "AI launcher" and "prompt-to-URL" actions only.
4. Mobile/iPadOS support.

4. Users & Key Use Cases (from observed real-world setups)

Persona	Observed usage (sourced from community research)
Office / Excel power user	Top-level Copy, Paste, Clipboard history, Screenshot; Excel folder (Create Table, Create Pivot, monthly-report macro opening 5 URLs + a workbook); window-management folder; company-app folder.
Creative pro (Photoshop / Premiere / Resolve)	Per-app rings: brush size $\pm$, undo/redo history scrubbing via dial, Remove/Paste attributes, Nest, razor, workspace switching, export presets; adjustment sliders for brightness/contrast/exposure.
CAD drafter	Ring as app/file launcher (taskbar hidden); "keystroke-then-Enter" commands (AutoCAD R, O, Enter).
Developer / multi-PC user	Device/desktop switching, terminal commands via Run, AI chat launcher, scripts (AppleScript/PowerShell) on bubbles.
Meetings user	Teams/Zoom ring: mute mic, share screen, chat, reactions, camera toggle.

5. Competitive Landscape

Product	Platform	Strengths	Weaknesses our product exploits
Logitech Actions Ring (reference)	Win 10+/macOS 13+	Polished drag-drop GUI; contextual plugins auto-creating app profiles; analog adjustment bars; macro (Smart Actions) integration; signed corporate-installable software; hardware trigger + haptics on MX Master 4	Requires Logitech hardware + internet + account; cold-start delay of 10–20 s after wake; fragile LogiPluginService (works-once-then-dies bugs); hard 8-slot / 1-level-folder cap; no disable toggle or game mode; heavy background processes; telemetry concerns
Kando (OSS)	Win/mac/Linux	Unlimited nesting; marking-menu gesture selection; themes; offline; free	Electron footprint; no analog adjustment bars; no plugin marketplace; config heavy

Radial (AppVerge)	macOS	Sub-menus, visual macro editor, shell scripts, Apple Shortcuts, preset marketplace	Mac-only, paid
Pie Menu / Pieoneer / Cirmenu	macOS	Menu-bar harvesting (Pie Menu); polish (Pieoneer)	Narrow action types; Mac-only
AutoHotPie / AHK Radial / GestureSign	Windows	Free, fully scriptable	AHK runtime often banned in corporate environments; no polished GUI
Elgato Virtual Stream Deck	Win/mac	More slots/layouts, Shortcuts/Keyboard Maestro macros	Requires Stream Deck hardware; grid not radial

Positioning: "The Logitech Actions Ring for every mouse — instant, offline, and unlimited." Combine Logitech's polish + analog adjustments + contextual plugins with Kando's openness, depth and offline-first stance.

6. Detailed Feature Specification

Priorities: **P0** = MVP, must ship first release; **P1** = v1.x fast-follow; **P2** = v2 / stretch. Requirements marked **[Δ]** intentionally exceed Logitech's implementation to answer a documented community complaint.

F1 — Ring Overlay (core surface)

- F1.1** **P0** On trigger, render a radial overlay of round action "bubbles" **centered on the current cursor position**, on any monitor, above all windows (topmost, click-through-exempt). Bubbles are spaced equidistantly around the cursor.
- F1.2** **P0** Default bubble count is **8**. **[Δ]** Count is configurable per profile from 2–12 (Logitech hard-caps at 8; "more slots" is community request #4).
- F1.3** **P0** Each bubble shows an icon and, on hover, a text label. Hovered bubble visibly highlights (scale + accent color).
- F1.4** **P0** The ring stays open until the user selects an action, presses the trigger again, presses `Esc`, or clicks outside the ring. Selecting fires the action and closes the ring (folders excepted, F4).
- F1.5** **P0** Two activation gestures, matching Logitech's: **(a) click-release** — tap trigger, ring opens and persists, click a bubble; **(b) press-hold-release** — hold trigger, move to bubble, release to fire. Both work without configuration.
- F1.6** **P1** **[Δ] Marking-menu quick select** (Kando-style): with gesture (b), a fast directional flick selects the bubble in that direction even before the overlay finishes rendering — expert users never wait for the UI.
- F1.7** **P0** Three ring sizes — Small / Medium (default) / Large — set globally in Settings (parity with Options+ v2.4). **P1** plus a continuous scale slider (75–150%).
- F1.8** **P0** Overlay opacity configurable (Logitech tuned opacity for readability in v1.95; ProVideo Coalition asked for a contrast setting). Default ≥ 95% with a high-contrast theme option.
- F1.9** **P1** Open/close animation (~120 ms scale/fade), disableable ("reduce motion" respects OS accessibility setting).
- F1.10** **P0** Multi-monitor and per-monitor DPI aware; ring must render fully on-screen (clamp position when triggered near screen edges — cursor-relative but never clipped).
- F1.11** **P1** Behavior over fullscreen/exclusive apps is explicit: detect exclusive-fullscreen and either suppress the ring (default, prevents game disruption) or render borderless-topmost, per-profile setting. (Logitech's ring silently fails over fullscreen apps — community bug \$5.)
- F1.12** **P2** Theming: user-selectable themes (light/dark/auto + accent), custom theme files (colors, bubble shape, fonts) à la Kando.

F2 — Trigger System

- F2.1** **P0** Trigger sources: (a) any mouse button (including side/extra buttons XBUTTON1/2, middle click) via low-level mouse hook; (b) any keyboard hotkey/combination; (c) **P1** mouse gesture (e.g. hold right button + flick) and double-modifier taps.
- F2.2** **P0** "Show Action Ring" is a mappable action; multiple simultaneous triggers allowed (e.g. mouse button AND hotkey).

F2.3 P0 When a mouse button is consumed as the trigger, its original event must be swallowed (no click leaks to the app underneath), and un-mapping must fully restore default behavior.

F2.4 P0 **[Δ] True disable:** a first-class global toggle (tray menu + hotkey + a "Toggle Action Ring" assignable action) that fully disarms all triggers. Logitech's only mechanism is remapping to "Do nothing", which still pops an empty ring for some users — community request #1.

F2.5 P1 **[Δ] Game mode:** auto-disable when a process on a user-managed blocklist (or any exclusive-fullscreen app) is foreground; tray icon reflects the armed/disarmed state — community request #2.

F2.6 P1 Debounce/sensitivity: configurable minimum hold time (0–500 ms) before the ring opens, to suppress accidental triggers (the MX Master 4's #1 hardware complaint; we solve it in software).

F3 — Actions (assignable to any bubble)

F3.1 P0 **Keystroke:** send any single shortcut (modifiers + key), recorded via a key-capture field. Must support media keys and OS keys.

F3.2 P0 **[Δ] Keystroke sequence:** ordered multi-chord sequences in one action (e.g. Excel `Alt+A`, then `C`; AutoCAD `R`, `0`, `Enter`) with per-step delay. Logitech cannot do this in one action — community request #6.

F3.3 P0 **Open application** (launch or focus-if-running — "switch to" semantics like AppleInsider's OmniFocus example).

F3.4 P0 **Open folder / file / URL.** Folder-open must be instant for network paths (spawn `explorer.exe <path>` asynchronously — Logitech's implementation blocks up to 20 s on network drives, community bug \$5).

F3.5 P0 **Run command:** arbitrary command line with arguments; optional "run hidden" and "run as admin" flags; P1 inline script actions (PowerShell/AppleScript/bash) with the script stored in the profile.

F3.6 P0 **System actions** (parity list from Options+): copy, paste, undo, redo, cut; screenshot (region/window/full); volume up/down/mute; media play-pause/next/previous; brightness $\pm$; lock workstation; sleep displays; show desktop; window snap left/right/maximize/minimize; new note; emoji picker; clipboard-history picker; app switcher; virtual-desktop next/previous; zoom in/out; search (OS search).

F3.7 P0 **Multi-Action (macro):** ordered list of any other actions with configurable inter-step delays and "type text" steps (parity with Smart Actions' core: "open multiple apps or websites with a single click... add a delay in between each action"). P1 conditions (if app running / if window title matches) and **[Δ]** a cancel mechanism (Esc aborts a running macro — community request #13).

F3.8 P1 **Text snippet:** paste a stored text block (Radial-style; not in Logitech's ring).

F3.9 P1 **AI launcher:** open a chosen AI chat (ChatGPT / Gemini / Perplexity / Copilot / custom URL), optionally passing the current selection or clipboard as URL-encoded prompt (parity with the "Explore AI" default bubble; superset via prompt passing).

F3.10 P1 **Switch profile** action (jump to a named profile or back to General — parity with v1.91) and **Do nothing** placeholder action.

F3.11 P2 **Device switch** (Easy-Switch equivalent): where device SDKs allow (e.g. Logitech devices via logi HID++, or generic Bluetooth profile switching), expose a channel-switch action; otherwise omit gracefully.

F3.12 P2 **[Δ] Dynamic bubbles:** bubble types whose contents are computed at open time — "open windows" ring (Alt-Tab-style, community request #7), "recent files", "clipboard history top 8".

F4 — Folders (nesting)

F4.1 P0 Any bubble can be a **Folder** containing up to 9 sub-bubbles (parity: $8 \times 9 = 72$ actions). Clicking a folder expands its sub-bubbles in place (ring morphs; back affordance returns to the top level).

F4.2 P1 Δ Nesting depth is configurable up to 3 levels (Logitech allows exactly 1; Kando proves deeper nesting works).

F4.3 P0 Δ Actions can be removed from folders, reordered, and moved between folders/top level in the editor (Logitech cannot delete-from-folder — community request #9).

F4.4 P0 Folders have their own name and icon, both editable.

F5 — Adjustment Bar (analog actions)

F5.1 P0 An **adjustment action** binds a continuous parameter (volume, brightness, zoom, scroll, or a plugin parameter such as brush size / exposure) to relative input. While hovering the bubble, the **scroll wheel** adjusts the value.

F5.2 P0 Click-and-drag on an adjustment bubble reveals a horizontal **adjustment bar**; dragging right/left changes the value continuously (full parity with Logitech's drag behavior).

F5.3 P0 Δ The current **numeric value is displayed** during adjustment (e.g. "68%") — Logitech shows none; community request #8.

F5.4 P1 Per-action adjustment **sensitivity** setting (parity with v1.98) and optional snap/step values.

F5.5 P1 Generic "keystroke pair" adjustment: map wheel-up/wheel-down while hovering to two shortcuts (e.g. [/] for brush size) so any app gains analog control without a plugin.

F6 — Contextual Per-App Profiles

F6.1 P0 The ring detects the **foreground application at trigger time** and shows that app's profile if one exists, else the **General profile**. Detection by process name; P1 also window-title regex (for browser-based apps) and URL matching via browser extensions P2.

F6.2 P0 Unlimited named profiles; each holds its own ring layout (bubbles, folders, count). Profiles listed as tabs in the editor (parity with Options+ layout).

F6.3 P0 Create a profile from: blank, duplicate of General, or a bundled preset (see F8). "Add application" flow lists running apps + installed apps + browse-to-exe.

F6.4 P1 Profile **import/export** as a single shareable file (parity with v2.1 import; superset: export too, enabling community sharing without a marketplace).

F6.5 P1 Per-profile overrides: bubble count, ring size, fullscreen behavior, trigger hold-time.

F7 — Configuration & Editor UI

F7.1 P0 A settings/editor app with a live ring preview: **drag-and-drop** actions from a searchable, categorized picker (System / Apps / Macros / Plugins) onto bubbles; drag between bubbles to reorder; drag off to remove.

F7.2 P0 Per-bubble settings: rename, choose icon, configure action parameters.

F7.3 P0 **Icon system**: bundled icon set (≥ 300 glyphs, e.g. Lucide/Font Awesome Free); auto-extracted app icons from the target exe; user-supplied PNG/SVG with validation and a clear error message (Logitech

added an SVG-compatible popup in v2.3); **P1** icon editor (background color, glyph color, text label, size) and installable icon packs.

F7.4 **P1** Undo/redo in the editor; changes apply live (next ring open uses the new layout, no restart).

F7.5 **P0** First-run onboarding: 3-step tour (pick trigger → try the ring → open editor), and a default General profile that demonstrates value immediately (parity with Logitech's defaults: play/pause, screenshot, clipboard history, emoji, lock, file manager, AI launcher, settings).

F7.6 **P1** Settings search; keyboard-navigable editor; system tray icon with quick actions (enable/disable, open editor, switch profile, quit).

F8 — Plugins & Presets (app integrations)

F8.1 **P0** Ship **bundled preset profiles** for: Chrome/Edge/Firefox, Excel, Word, PowerPoint, Photoshop, Premiere Pro, Lightroom Classic, DaVinci Resolve, Figma (web+app), VS Code, Teams, Zoom, Slack, Spotify, File Explorer/Finder — built from documented shortcut sets, no host-app SDK needed. (Firefox preset is itself a fix: Logitech covers Chrome only.)

F8.2 **P1** **Plugin SDK v1 (declarative)**: a plugin = JSON manifest + icon assets defining actions (keystrokes/sequences/commands/adjustment pairs) and the app-match rule; installing a plugin auto-creates its contextual profile (parity with Logi Marketplace behavior).

F8.3 **P2** **Plugin SDK v2 (programmatic)**: local IPC (named pipe / WebSocket) so a plugin process can expose live actions and receive adjustment deltas (needed for true Photoshop-style parameter control à la Loupedeck). Sandboxed, permission-prompted.

F8.4 **P2** In-app **gallery** browsing a static, signed index (GitHub-hosted) of community presets/plugins/icon packs — works read-only, degrades gracefully offline. No account required, ever.

F9 — Storage, Sync & Offline Behavior

F9.1 **P0** All configuration is **local, human-readable JSON** under the user profile dir, written atomically; corrupted files are backed up and regenerated, never silently reset.

F9.2 **P0** **100% offline operation** of every core feature (ring, actions, editor, presets). This is the #3 community request and Logitech's single biggest flashpoint (their official statement: ring + Smart Actions "require an active internet connection").

F9.3 **P1** Automatic local config backups (last 10 versions) + one-click full backup/restore file.

F9.4 **P2** Optional folder-based sync (point the config at Dropbox/OneDrive/Syncthing) — no proprietary cloud.

F10 — Performance & Reliability (product-defining, not an afterthought)

F10.1 **P0** Trigger-to-visible-ring latency < **100 ms p99**, including the first invocation after boot/sleep (Logitech's 10–20 s cold start is the #1 community complaint; the overlay process must stay warm and pre-rendered).

F10.2 **P0** Idle footprint: ≤ 50 MB RAM, ≤ 0.1% CPU, zero network. Single background process (no separate updater/agent/broker trio — Options+ runs 3+ processes and is a documented battery drain).

F10.3 **P0** Action execution starts ≤ 50 ms after bubble selection; actions fired from the ring must be as fast as a direct button binding (a measured Logitech regression).

- F10.4** P0 Watchdog & crash recovery: if the overlay process dies, it restarts silently ≤ 2 s; hooks re-register after resume-from-sleep and after Explorer restarts. No "works once then dies until reboot" states (documented Options+ v2.4 bug).
- F10.5** P1 Diagnostics page: hook status, last-trigger latency, per-action success log; "restart engine" button that actually works (Options+'s restart hangs in a loop for many users).
- F10.6** P1 Known-conflict handling: detect focus-follows-mouse tools/password-manager overlays that inject ghost clicks (documented AutoRaise conflict) and warn with guidance rather than glitching.

F11 — Privacy, Security & Distribution

- F11.1** P0 **No telemetry by default**; optional anonymous usage stats strictly opt-in with a visible data list. No ads, no account, no bundled software (each of these is a documented Options+ grievance).
- F11.2** P0 Signed installer (MSI/MSIX + notarized DMG later); silent-install flags for corporate deployment; no admin needed to run.
- F11.3** P1 Minimal OS permissions, each requested lazily with an explanation (macOS: Accessibility only; never Screen Recording unless the screenshot action is used — Options+'s "crazy" permission sprawl is a documented IT blocker).
- F11.4** P1 Updates: manual check by default, optional auto-update; the updater must not run persistently.

F12 — Accessibility & Localization

- F12.1** P1 Full keyboard operation of the ring (arrow keys/numbers select bubbles, Enter fires, Esc closes); screen-reader labels on bubbles; respects OS reduce-motion and high-contrast modes.
- F12.2** P2 Localization framework + top-10 languages; correct shortcut handling across keyboard layouts (Logitech had to patch non-US layouts for its plugins).

7. Out-of-scope Reference: Logitech Features Intentionally Dropped

- **Hardware haptics** (hover/select bumps, 4 intensity levels) — device-firmware dependent; revisit with Windows 11 Advanced Haptics (P2 research task T9.4).
- **AI Prompt Builder** full clone (ChatGPT window with Rephrase/Summarize/Reply/Create Email recipes, tone/length dropdowns) — replaced by the simpler F3.9 AI launcher; a recipes-style prompt palette is a possible v3.
- **Logi Flow-style multi-computer control** — separate product territory.
- **Onboard device memory** — impossible without our own hardware; our answer is instant-start local software.

8. Non-Functional Requirements Summary

Attribute	Requirement
Latency	Trigger→ring < 100 ms p99 (incl. cold start); selection→action < 50 ms
Footprint	≤ 50 MB RAM idle; $\leq 0.1\%$ CPU idle; installer ≤ 40 MB (vs Options+ offline 750 MB)
Reliability	Crash-free sessions $\geq 99.5\%$; auto-recovery ≤ 2 s; survives sleep/wake, monitor changes, DPI changes

Compatibility	Windows 10 21H2+ / Windows 11 (x64 + ARM64 — Logitech shipped an ARM rendering bug; we test it); macOS 13+ (phase 2)
Privacy	Zero network calls unless gallery/update explicitly invoked; opt-in telemetry only
Input safety	Hooks never add measurable pointer latency (< 1 ms processing per event); clean unhook on exit

9. Success Metrics

- **Activation:** $\geq 60\%$ of installers trigger the ring ≥ 10 times in week 1 (onboarding quality).
- **Retention:** $\geq 40\%$ of week-1 actives still use the ring daily at day 30.
- **Depth:** median ≥ 2 customized bubbles and ≥ 1 app profile per retained user by day 30.
- **Performance honesty:** p99 cold-start invocation < 100 ms measured in the field (diagnostics opt-in), not lab-only.
- **Quality:** < 1% of sessions with a failed trigger (ring didn't appear) — the exact failure mode that dominates Logitech complaint threads.

10. Risks & Open Questions

#	Risk / question	Mitigation / decision needed
R1	Low-level mouse hooks are flagged by some anti-cheat systems	Game mode (F2.5) disarms hooks per-process; document clearly
R2	Framework choice: native (Win32/WinUI, ~10 MB, fastest) vs Electron/Tauri (faster UI dev, heavier)	Recommend: Tauri or native shell for the overlay + web-tech editor; overlay latency budget rules out Electron for the ring itself. Decide in T0.2
R3	Swallowing a mouse button globally can break edge cases (drag operations, RDP, virtualization)	Extensive input-matrix testing (T1.6); passthrough allowlist
R4	Trademark: "Actions Ring" is Logitech's marketing name	Ship under a distinct name/brand; "Action Ring" as working title only — legal review before launch
R5	Preset shortcut sets rot as host apps update (documented Logitech pain)	Presets are data-only JSON → hot-fixable via gallery; community PRs
R6	macOS Accessibility-permission friction	Phase 2; polished permission onboarding, single permission only
Q1	Monetization (free/OSS core + paid pro? one-time license?)	Open — does not block v1 build
Q2	Do we pursue Logitech-device deep integration (HID++ for Easy-Switch/haptics)?	Defer to P2 (F3.11); investigate legality/SDK access

11. Release Plan

Release	Scope	Exit criteria
M0 — Spike (wk 1–2)	Tech decisions + overlay/hook proofs of concept	<100 ms overlay POC on Win10/11 incl. ARM64
M1 — MVP (wk 3–8)	F1 core, F2.1–2.4, F3.1–3.7, F4.1/4.3/4.4, F6.1–6.3, F7 basic editor, F9.1–9.2, F10 targets	Daily-drivable; all P0s demoable; perf budgets green
M2 — v1.0 public (wk 9–14)	F5 adjustments, F7 full editor + onboarding + icons, F8.1 presets, F2.5–2.6, installer/signing, diagnostics	All P0 complete, crash-free $\geq$ 99.5% in beta
M3 — v1.x (wk 15–22)	P1 set: marking menus, deeper nesting, profile import/export, plugin SDK v1, snippets, AI launcher, accessibility	P1 complete; community preset sharing live
M4 — v2 (later)	macOS port, plugin SDK v2, dynamic bubbles, gallery, themes, haptics research	—

12. Work Breakdown — Tasks

Effort: $S \leq 1$ dev-day · $M \leq 3$ · $L \leq 7$ · $XL > 7$. Dependencies reference task IDs.

Phase 0 — Foundations & spikes (M0)

ID	Task	Effort	Deps
T0.1	Write engineering design doc: process model (background engine + overlay + editor), config schema v1, IPC choice	M	—
T0.2	Framework spike: measure overlay show-latency for native Win32/D2D vs Tauri vs Electron; pick stack (decides R2)	M	—
T0.3	Low-level input POC: global mouse+keyboard hooks (WH_MOUSE_LL / WH_KEYBOARD_LL or Raw Input), swallow XBUTTONDOWN events, re-inject when unarmed	M	—
T0.4	Input injection POC: SendInput for chords, sequences, unicode text; verify against elevated windows (UIPI limits documented)	S	—
T0.5	Foreground-app detection POC: process name + window title on trigger, incl. UWP apps; measure cost	S	—
T0.6	CI setup: build, unit tests, signed nightly artifact; ARM64 build target from day one	M	T0.2

Phase 1 — MVP ring engine (M1)

ID	Task	Effort	Deps
T1.1	Background engine service: startup, tray icon, config load/watch, atomic JSON persistence (F9.1)	L	T0.1
T1.2	Trigger system: mouse-button + hotkey triggers, arm/disarm, global toggle, event swallowing (F2.1–2.4)	L	T0.3, T1.1
T1.3	Overlay renderer: pre-warmed topmost window, 8-bubble radial layout at cursor, hover highlight, labels, sizes S/M/L, DPI/multi-monitor clamping (F1.1–1.5, 1.7, 1.10)	XL	T0.2

T1.4	Both activation gestures: click-release and press-hold-release selection (F1.5)	M	T1.2, T1.3
T1.5	Action framework + executors: keystroke, keystroke sequence, open app/folder/file/URL, run command, do-nothing (F3.1–3.5, 3.10)	L	T0.4, T1.1
T1.6	Input-safety test matrix: drag ops, RDP, VMs, elevated apps, games, multiple mice; fix leaks (R3)	M	T1.2
T1.7	System actions pack: clipboard, media, volume, screenshot, window management, lock, emoji, clipboard history (F3.6)	L	T1.5
T1.8	Multi-Action macro engine with delays + type-text steps (F3.7)	M	T1.5
T1.9	Folders: data model, in-ring expansion, back navigation (F4.1, 4.4)	M	T1.3
T1.10	Contextual profiles: General + per-app by process name, profile resolution at trigger time (F6.1–6.3)	M	T0.5, T1.1
T1.11	Minimal editor v0: profile list, bubble grid, action picker (list-based), rename/icon-from-exe (F7.1–7.2 subset)	L	T1.1
T1.12	Performance harness: automated trigger→pixel latency measurement; cold-start test after simulated sleep; RAM/CPU budget gate in CI (F10.1–10.3)	M	T1.3
T1.13	Watchdog + resume-from-sleep hook re-registration + crash recovery (F10.4)	M	T1.1

Phase 2 — v1.0 polish (M2)

ID	Task	Effort	Deps
T2.1	Adjustment actions: scroll-wheel-on-hover, click-drag adjustment bar, live numeric value display, volume/brightness/zoom built-ins (F5.1–5.3)	L	T1.3, T1.7
T2.2	Full editor: drag-and-drop from categorized searchable picker, drag-reorder/remove, folder editing incl. remove-from-folder (F7.1, F4.3)	XL	T1.11
T2.3	Icon system: bundled glyph set, exe icon extraction, PNG/SVG import with validation (F7.3)	M	T2.2
T2.4	Onboarding flow + default General profile (F7.5)	M	T2.2
T2.5	Game mode + fullscreen detection + per-profile fullscreen policy + trigger hold-time debounce (F2.5–2.6, F1.11)	M	T1.2, T1.10
T2.6	Bundled presets: author + verify 15 app preset profiles (F8.1)	L	T1.10
T2.7	Opacity/high-contrast setting, open/close animation with reduce-motion (F1.8–1.9)	S	T1.3
T2.8	Diagnostics page: hook status, latency stats, action log, engine restart (F10.5)	M	T1.13
T2.9	Signed installer (MSI/MSIX), silent-install flags, autostart registration, clean uninstall (F11.2)	M	T0.6
T2.10	Beta program: crash reporting (opt-in), feedback channel, fix cycle to ≥ 99.5% crash-free	L	all P0
T2.11	ARM64 validation pass (Logitech's black-rectangle bug class) + Win10 21H2 compatibility pass	S	T2.10

Phase 3 — v1.x differentiators (M3)

ID	Task	Effort	Deps
T3.1	Marking-menu flick selection (F1.6)	M	T1.4
T3.2	Configurable bubble count 2–12 + per-profile overrides (F1.2, F6.5)	M	T2.2
T3.3	Nesting to 3 levels (F4.2)	M	T1.9

T3.4	Profile import/export file format + UI (F6.4)	M	T2.2
T3.5	Plugin SDK v1: manifest schema, loader, auto-profile creation, docs + 2 sample plugins (F8.2)	L	T2.6
T3.6	Text snippets, AI launcher with selection/clipboard prompt passing, switch-profile action (F3.8–3.10)	M	T1.5
T3.7	Macro conditions + Esc-cancels-macro (F3.7 ext)	M	T1.8
T3.8	Keystroke-pair generic adjustments + per-action sensitivity (F5.4–5.5)	M	T2.1
T3.9	Accessibility: keyboard ring navigation, screen-reader labels, high-contrast audit (F12.1)	M	T2.7
T3.10	Local config backups + restore (F9.3); conflict-tool detection warnings (F10.6)	S	T1.1
T3.11	Window-title/regex app matching (F6.1 ext)	S	T1.10

Phase 4 — v2 and beyond (M4)

ID	Task	Effort	Deps
T4.1	macOS port: NSEvent taps, Accessibility onboarding, notarized DMG (single permission, F11.3)	XL	M2
T4.2	Plugin SDK v2: IPC protocol, sandboxing, permission prompts; Photoshop/OBS reference plugins (F8.3)	XL	T3.5
T4.3	Dynamic bubbles: open-windows ring, recent files, clipboard top-8 (F3.12)	L	T3.2
T4.4	Community gallery: signed static index, in-app browse/install of presets/plugins/icon packs (F8.4)	L	T3.4, T3.5
T4.5	Theme engine + theme file format (F1.12)	M	T2.7
T4.6	Folder-based sync support + docs (F9.4)	S	T3.10
T4.7	Research: Windows 11 Advanced Haptics for compatible mice; HID++ Easy-Switch feasibility (F3.11, Q2)	M	—
T4.8	Localization framework + top-10 languages; layout-aware shortcut handling (F12.2)	L	T3.9

13. Appendix — Source Index

Full cited research (every fact with URL) lives in `research/01-official-logitech-sources.md`, `research/02-press-and-reviews.md`, `research/03-community-and-alternatives.md`. Primary hubs:

- Official: support.logi.com article 17844647906967 (Actions Ring FAQ); Options+ Release Notes (article 1500005516462); logitech.com/software/actions-ring; news.logitech.com press releases (MX Creative Console 2024, MX Master 4 2025, Signature Comfort Plus 2026); marketplace.logi.com; official YouTube tutorials (8 videos).
- Press: TechRadar, AppleInsider, MacStories, Macworld, Engadget, MakeUseOf, XDA, Forbes, ProVideo Coalition, Tom's Hardware, Pocket-lint, Windows Central, PCWorld, The Verge, How-To Geek, Thurrott, iMore, TrustedReviews, Galaxus, RTINGS, futurework.blog, dataholic.de.
- Community: r/logitech threads (1op07at, 1qp74ri, 1muj00z, 1nkgg4w, 1otoj7o, 1ougex2, 1nvl6sp, 1od4nn1, 1sjpu5c, 1ppkk0m, 1ogguwd, 1oxt5pu, 1plhned, 1nu64c3, 1nugle4, 1ffxjl7, 174mq8u), r/macapps 1j3vd7e, AutoHotkey forum t=139245, folivora.ai community, audiosex.pro, Hacker News item 47369819, hub.sync.logitech.com offline manual.
- Competitors: kando.menu, radial.appverge.net, pie-menu.com, AutoHotPie, GestureSign, Elgato Virtual Stream Deck coverage.